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College of Computer Science and Information Technology
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DEPARTMENT OF COMPUTER ENGINEERING

ARTI 406 – Image Processing

STUDENT PROCEDURAL MANUAL
Lab 10

2025 - 2026

IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

Lab 10 – Answers: Image Segmentation and Feature Extraction

Objective: Detect edges with Canny, detect corners with Harris, and segment an image with Otsu's thresholding.

Task 1: Canny Edge Detection

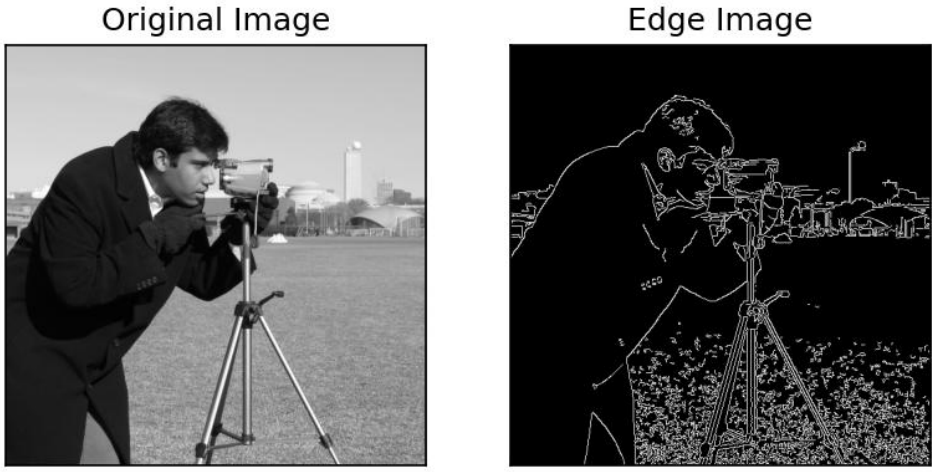


Figure 1: Camera image and its Canny edges (T_low = 100, T_high = 200)

Result: Canny gives thin, connected edges around the man, the camera and the tripod. The smooth sky gives no edges, while the rough grass gives many small edges.

Task 2: Harris Corner Detection

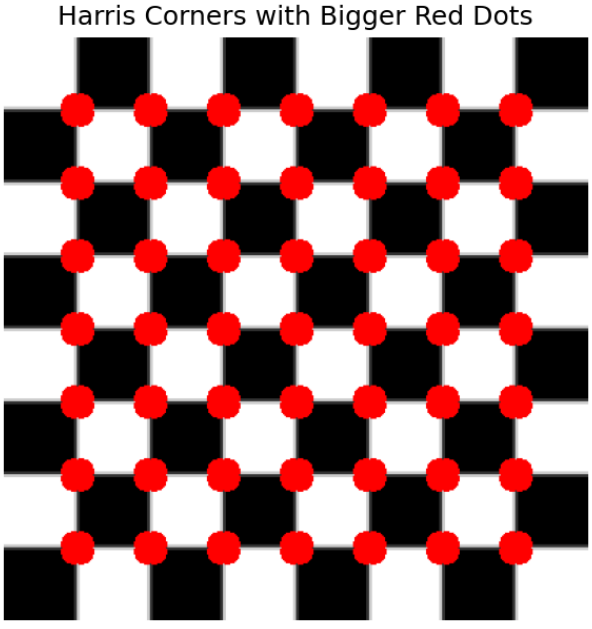


Figure 2: Harris corners on the checkerboard (red dots)

Result: Every inner point where four squares meet (49 points) is marked as a corner. Straight edges and flat squares are not marked, because a corner changes in two directions.

Q1: Otsu's Thresholding

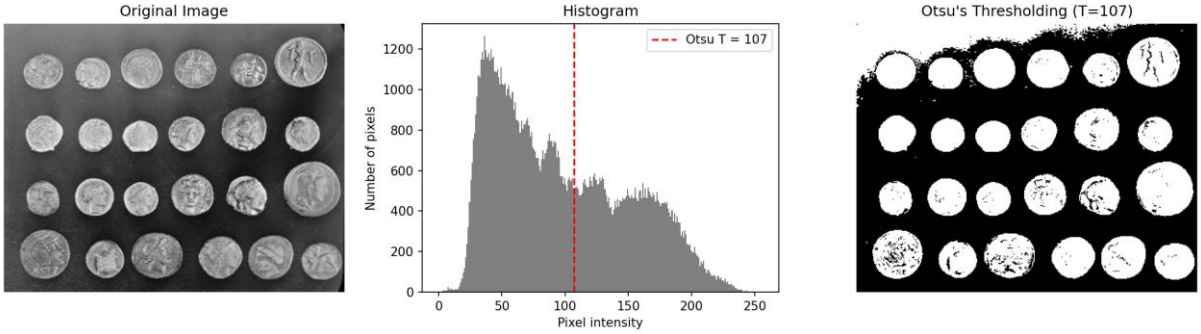


Figure 3: Original coins image, its histogram, and the result of Otsu's thresholding

Otsu threshold: T = 107 (found automatically).

Histogram: It has two main peaks: dark background (around 40) and bright coins (around 160). Otsu put T between them, where the variance between the two classes is largest.

Result: The coins become white and the background black. Compared with the fixed T = 127, Otsu gives fuller coins with fewer holes (38.8% white pixels instead of 29.6%). Some noise appears in the top-left corner, because that part of the background is brighter (uneven lighting).